Spatial Filtering

- Many image enhancement techniques are based on spatial operations that are performed on the local neighbourhood of the input pixels.

Spatial Filtering

Image Smoothing | Image Sharpening

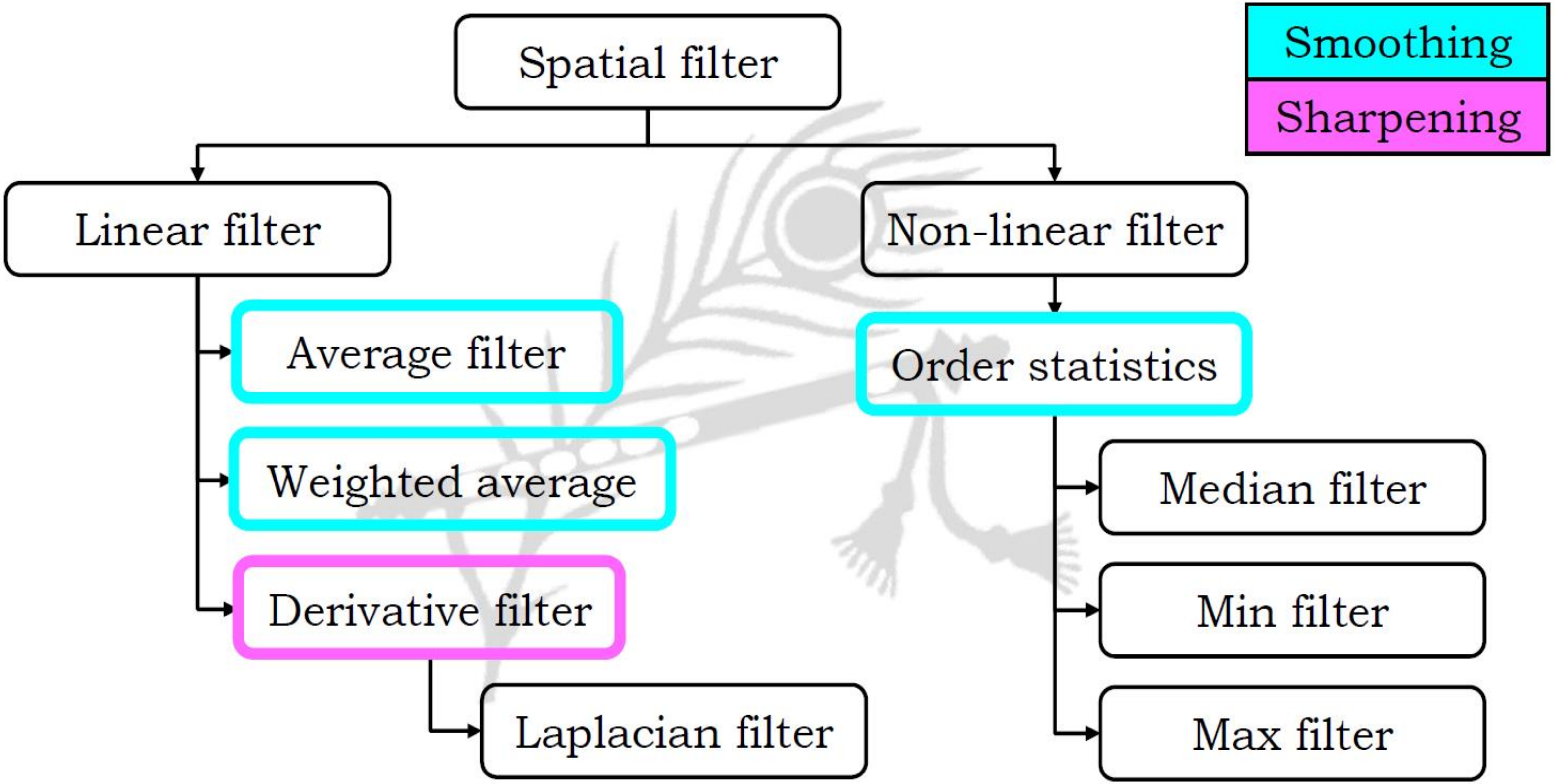
- Smoothing operation is used priminaly to reduce the effect of noise and to blur the false contours the may be present in digital image.
- These unwanted effects may be due to
  - → detection & recording error
  - → transmission channel noise
  - → digitization errors

Tr

Spatial Filtering

- Enhancement techniques based on the neighbours of pixel of an image are often referred to as Spatial filtering
- Filter term in "Digital image processing" is referred to the sub-image
- Sub-image is also termed as mask, kernel, template, or window
- The value in a filter sub-image are referred as coefficients, rather than pixels

Spatial Filtering

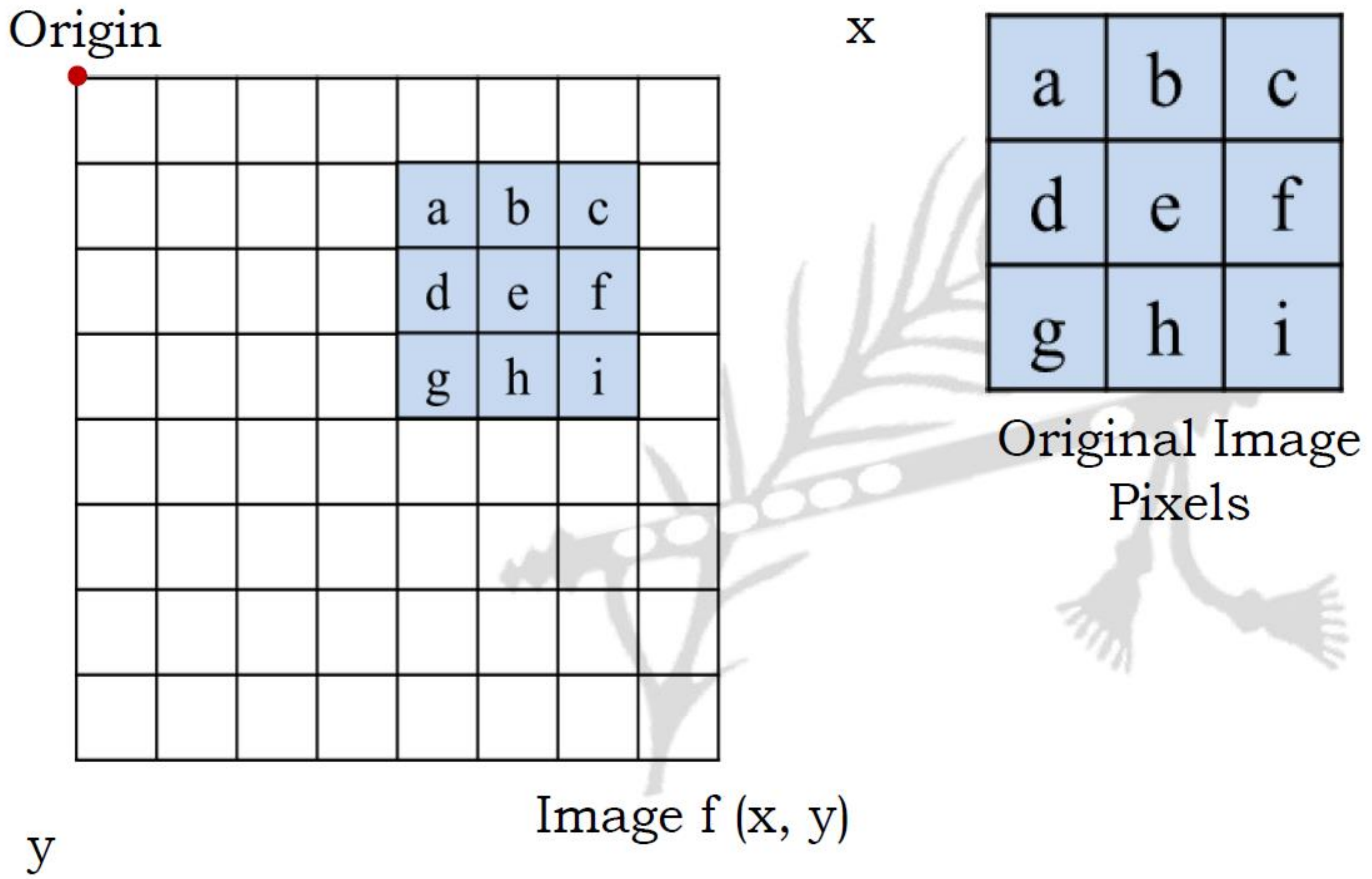


Mechanics of Spatial filtering

- The process consists simply of moving the filter mask from point to point in an image
- At each point (x, y), the response of the filter at that point is calculated using a predefined relationship

$$R = w_1 z_1 + w_2 z_2 + \ldots + w_{mn} z_{mn}$$
$$= \sum_{i=i}^{mn} w_i z_i$$

Mechanics of Spatial filtering



Mechanics of Spatial filtering

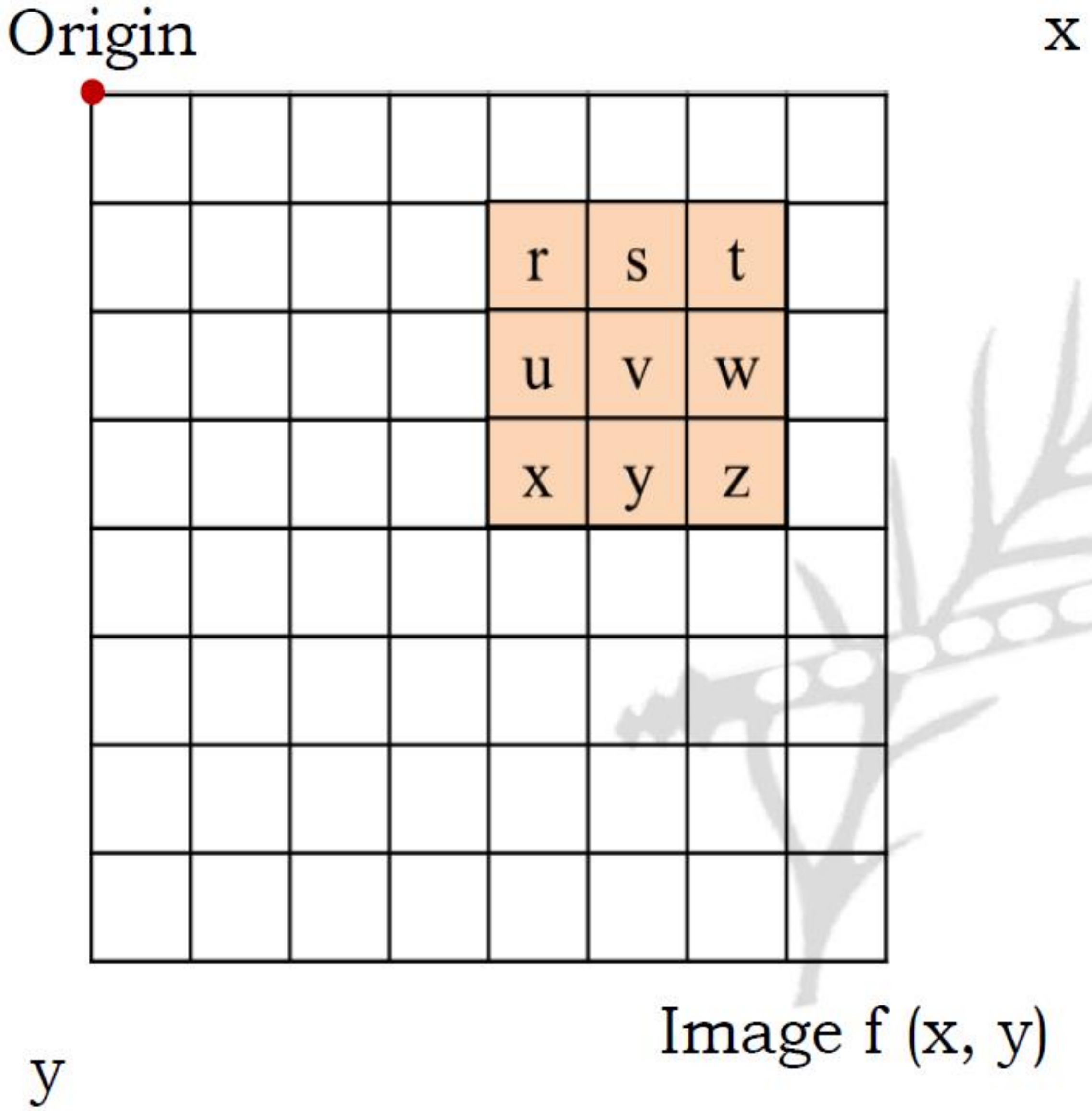


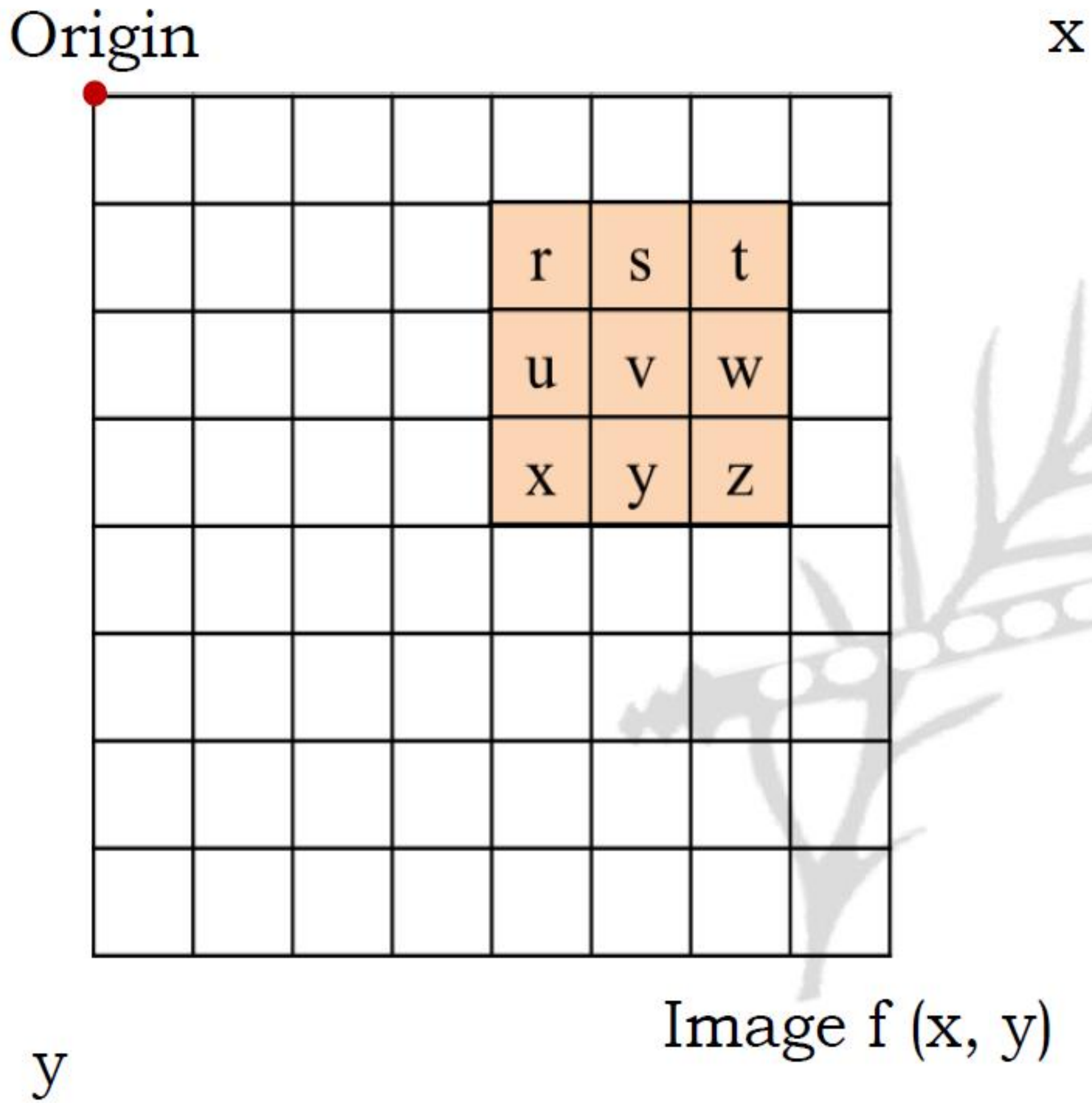
Image f (x, y)

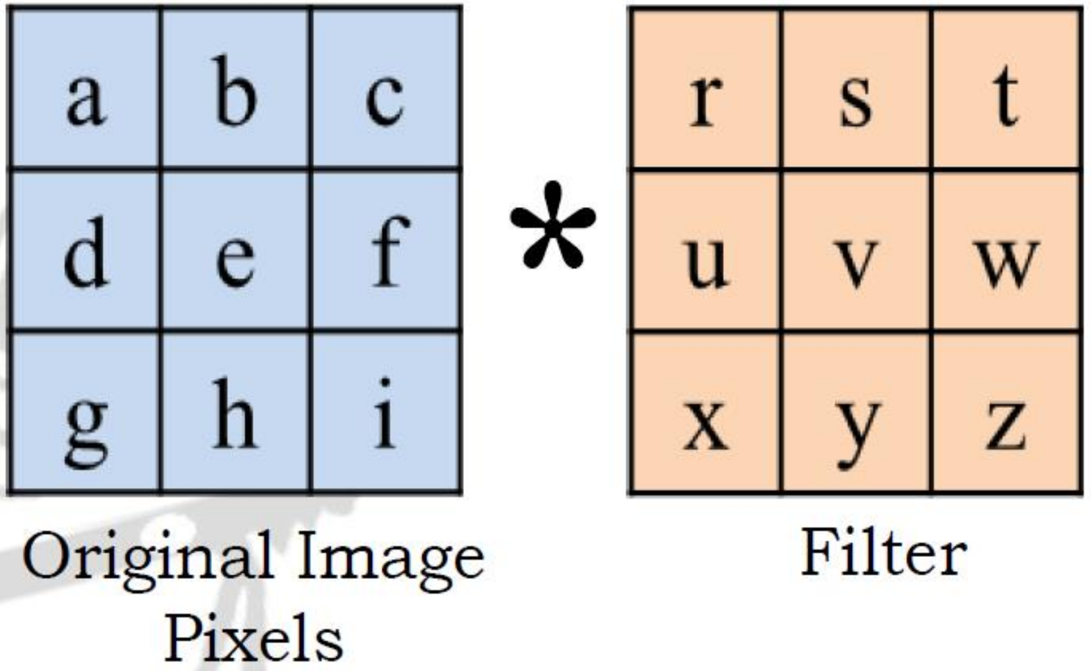
$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} * \begin{bmatrix} r & s & t \\ u & v & w \\ x & y & z \end{bmatrix}$$

Original Image Pixels

Filter

Mechanics of Spatial filtering





$$e_{processed} = r*a + s*b + t*c + u*d + v*e + w*f + x*g + y*h + z*i$$

The above is repeated for every pixel in the the original image to generate the filtered image

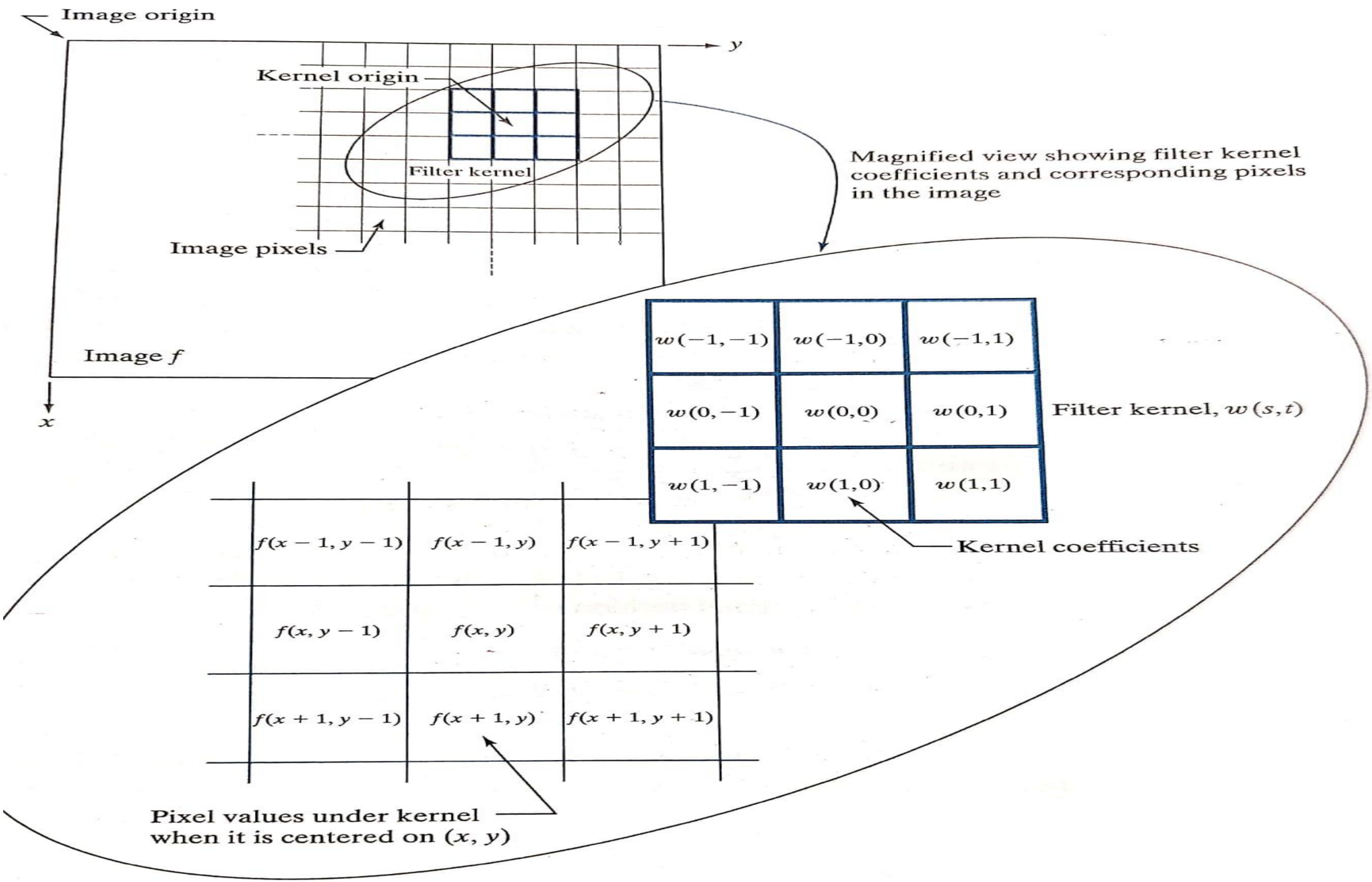


Image origin
y
Kernel origin
Filter kernel
Image pixels
Image f
x
Magnified view showing filter kernel coefficients and corresponding pixels in the image
w(−1,−1)
w(−1,0)
w(−1,1)
w(0,−1)
w(0,0)
w(0,1)
w(1,−1)
w(1,0)
w(1,1)
Filter kernel, w(s,t)
Kernel coefficients
f(x − 1, y − 1)
f(x − 1, y)
f(x − 1, y + 1)
f(x, y − 1)
f(x, y)
f(x, y + 1)
f(x + 1, y − 1)
f(x + 1, y)
f(x + 1, y + 1)
Pixel values under kernel when it is centered on (x, y)

In general, linear filtering of an image f of size M x N with a filter mask of size m x n is given by the expression:

$$g(x,y)=\sum_{s=-a}^{a}\sum_{t=-b}^{b}w(s,t)f(x+s,y+t)$$

Smoothing Spatial Filters

- Used for blurring and for noise reduction
- Blurring is used in preprocessing steps, such as
  - removal of small details from an image prior to object extraction
  - bridging of small gaps in lines or curves
- Noise reduction can be accomplished by blurring with a linear filter and also by a nonlinear filter
- There are 2 types of smoothing spatial filters
  - Linear filters
  - Order statistics filters

Smoothing Linear Filters

- Output is simply the average of the pixels contained in the neighborhood of the filter mask.
- Also Called **Averaging filters** or **Lowpass filters**.
- Replacing the value of every pixel in an image by the average of the gray levels in the neighborhood will reduce the "sharp" transitions in gray levels.
- Sharp transitions
  - **random noise in the image**
  - **edges of objects in the image**

Thus, smoothing can reduce noises (desirable) and blur edges (undesirable)

Linear filters → Average filter

- The general implementation for filtering an M x N image with a average/weighted average filter of size m x n is given by the expression

$$g(x,y)=\frac{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)f(x+s,y+t)}{\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)}$$

- Output is simply the average of the pixels contained in the neighbourhood of the filter mask
- Also called averaging filters or Lowpass filters and is useful for highlighting gross detail

$$\frac{1}{9}\times\begin{array}{|c|c|c|}\hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline\end{array}$$

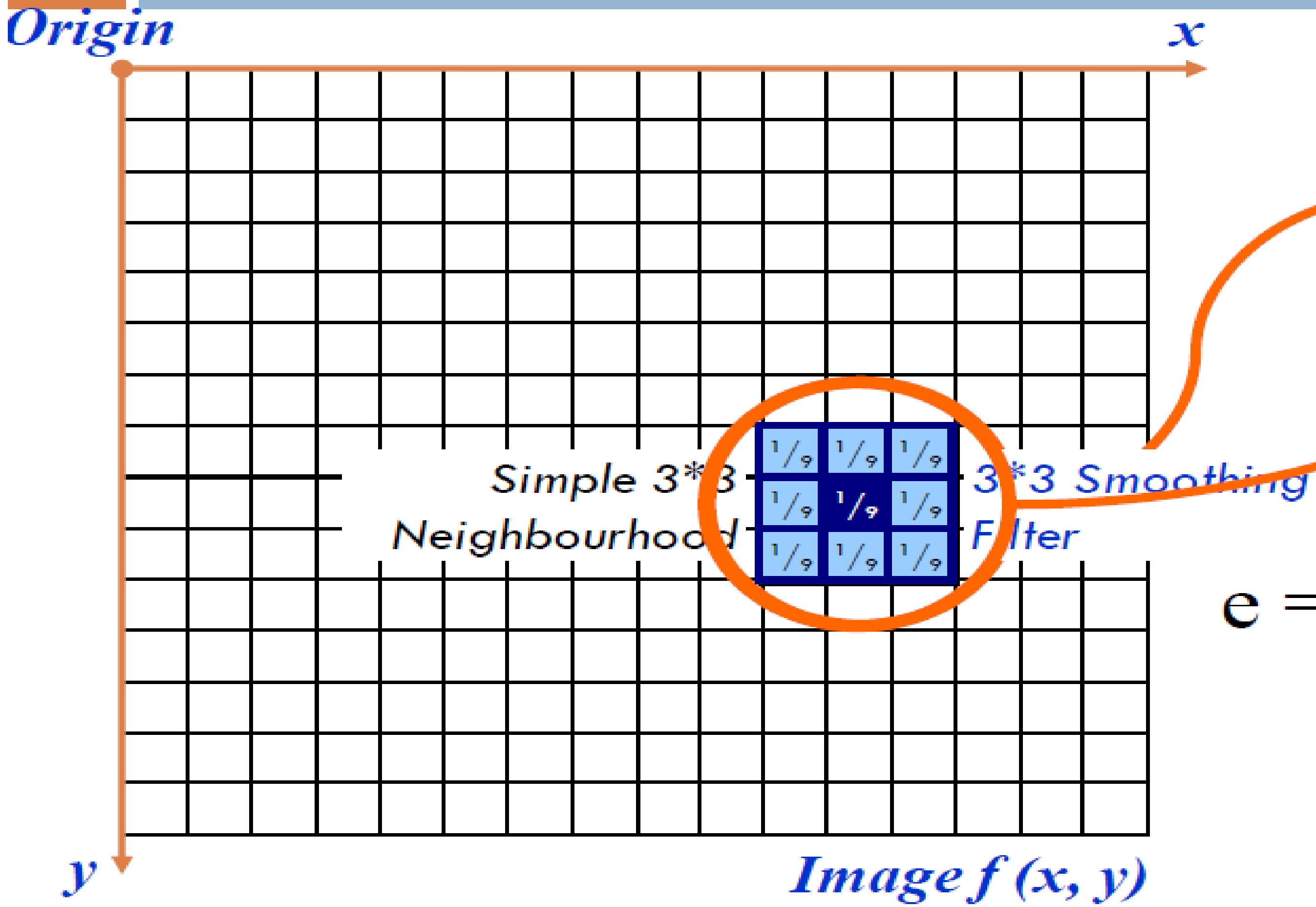
Smoothing Linear Filters …

- One of the simplest spatial filtering operations we can perform is a smoothing operation
  - Simply average all of the pixels in a neighbourhood around a central value
  - Especially useful in removing noise from images
  - Also useful for highlighting gross detail

$^1/_9$	$^1/_9$	$^1/_9$
$^1/_9$	$^1/_9$	$^1/_9$
$^1/_9$	$^1/_9$	$^1/_9$

Simple averaging filter

Smoothing Linear Filters …



104	100	108
99	106	98
95	90	85

Original Image Pixels

*

$^1/_9$	$^1/_9$	$^1/_9$
$^1/_9$	$^1/_9$	$^1/_9$
$^1/_9$	$^1/_9$	$^1/_9$

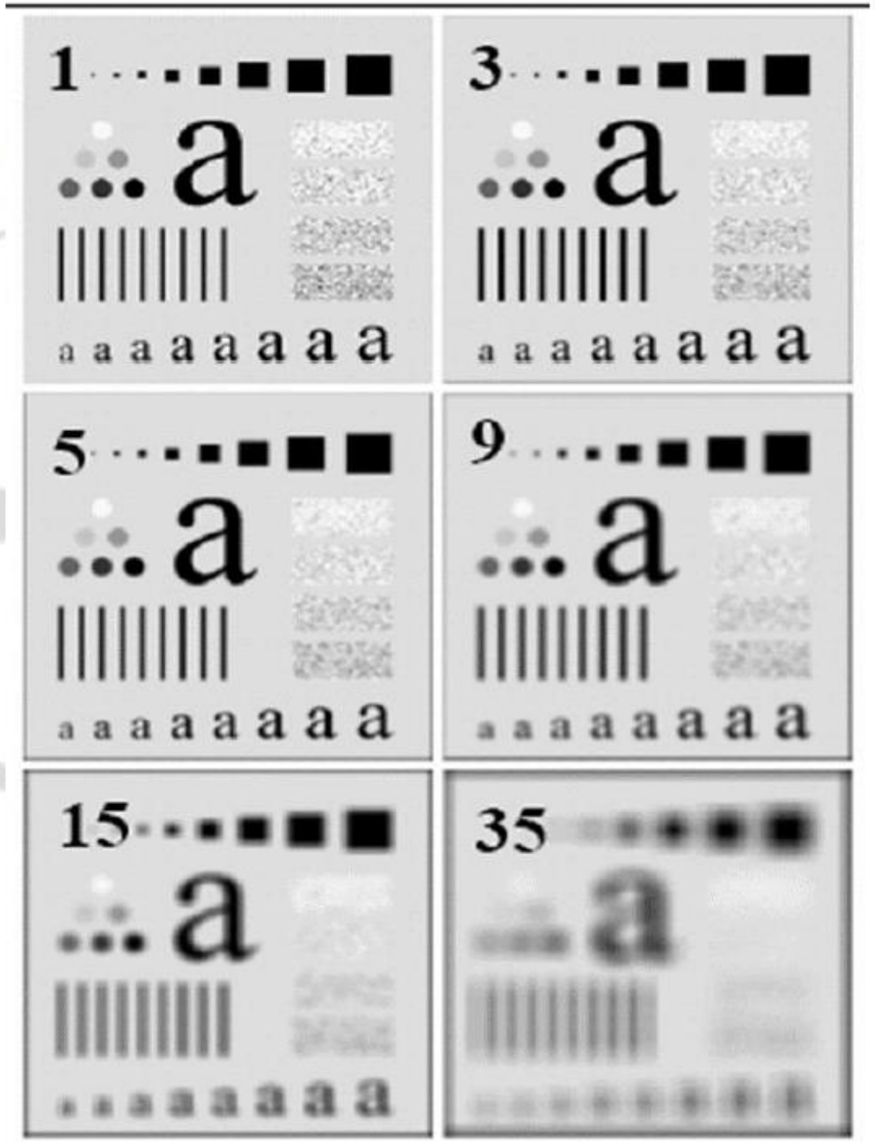
Filter

$$
\begin{aligned}
e = {} & {}^1/_9{*}106 + \\
& {}^1/_9{*}104 + {}^1/_9{*}100 + {}^1/_9{*}108 + \\
& {}^1/_9{*}99 + {}^1/_9{*}98 + \\
& {}^1/_9{*}95 + {}^1/_9{*}90 + {}^1/_9{*}85 \\
= {} & 98.3333
\end{aligned}
$$

The above is repeated for every pixel in the original image to generate the smoothed image

Linear filters → Average filter (Results)

- Original image of 500 x 500 pixels
- Results of smoothing with averaging filter masks of size m = 3, 5, 9, 15 and 35
- Note:
  - big mask is used to eliminate small objects from an image



Trade-off with an Average Filter

- It effectively reduces noise, it also blurs image details and edges significantly, especially when using a large filter window size.

- That means there's a balance between noise reduction and preserving sharpness in the image, where a larger filter window leads to more smoothing.

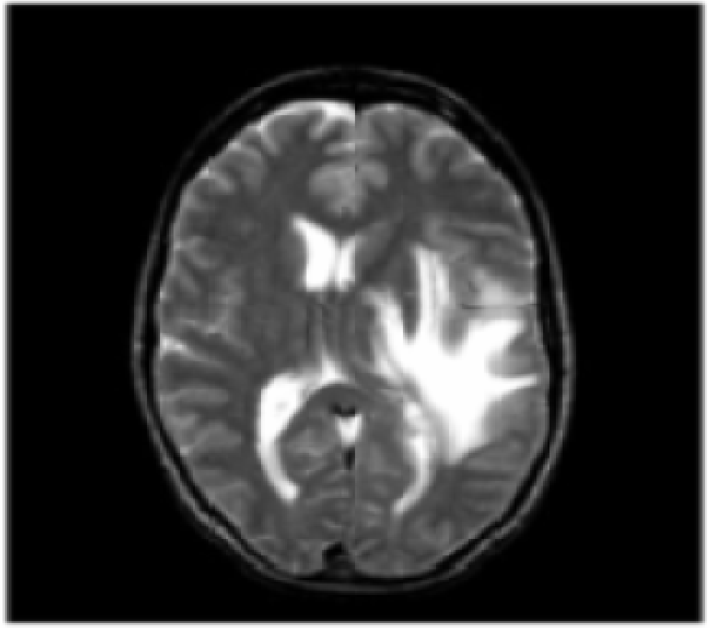
An average filter reduces noise in image processing by replacing each pixel value with the average of its surrounding pixels within a defined neighborhood, effectively smoothing out random fluctuations caused by noise while preserving the overall image features, as the noise tends to cancel itself out when averaged across multiple pixels; this is particularly effective for Gaussian noise where the noise values have a zero mean.

Linear filters → Weighted average filter

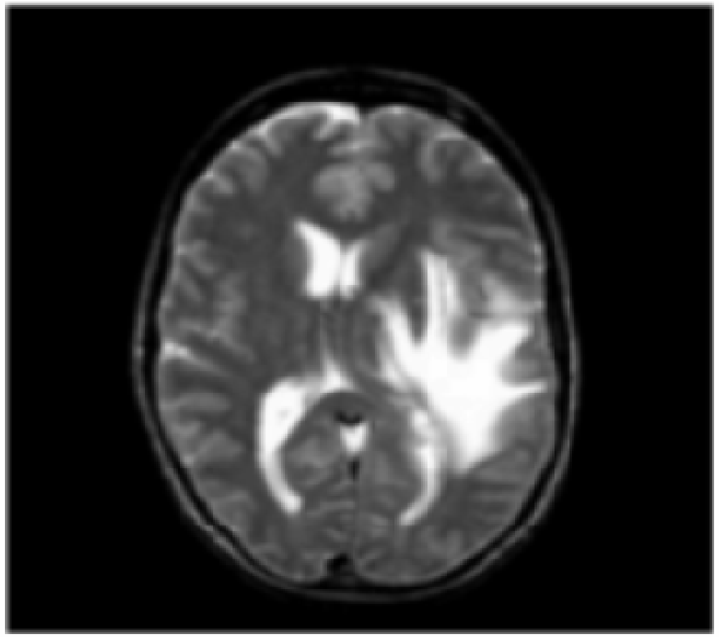
- More effective smoothing filters can be generated by allowing different pixels in the neighbourhood having different weights in the averaging function
- Pixels closer to the central pixel are more important
- Often referred to as a weighted averaging

$$\frac{1}{16}\begin{array}{|c|c|c|}\hline 1 & 2 & 1 \\ \hline 2 & 4 & 2 \\ \hline 1 & 2 & 1 \\ \hline \end{array}$$

Smoothing Linear Filters: Example



1/16	2/16	1/16
2/16	4/16	2/16
1/16	2/16	1/16



Before smoothing

After smoothing

Non-linear filters → Order statistics filters

- Order-statistics filters are nonlinear spatial filters whose response is based on ordering (ranking) the pixels contained in the image area encompassed by the filter,
- Types:
  - median filter : $R = \text{median}\{z_k \mid k = 1, 2, \ldots, n \times n\}$
  - max filter : $R = \max\{z_k \mid k = 1, 2, \ldots, n \times n\}$
  - min filter : $R = \min\{z_k \mid k = 1, 2, \ldots, n \times n\}$

Non-linear filters → Order statistics filters → Median filter

- Sort the values of the pixel in the region
- In the M x N mask, the median is the middle element
- Eg:

10	15	20
20	100	20
20	20	25

Solution:

- 10, 15, 20, 20, 20, 20, 20, 25, 100

Non-linear filters → Order statistics filters → Min filter

- Sort the values of the pixel in the region
- In the M x N mask, the min is the first element
- Eg:

10	15	20
20	100	20
20	20	25

Solution:

- 10, 15, 20, 20, 20, 20, 20, 25, 100

Non-linear filters → Order statistics filters → Max filter

- Sort the values of the pixel in the region
- In the M x N mask, the max is the last element
- Eg:

10	15	20
20	100	20
20	20	25

Solution:

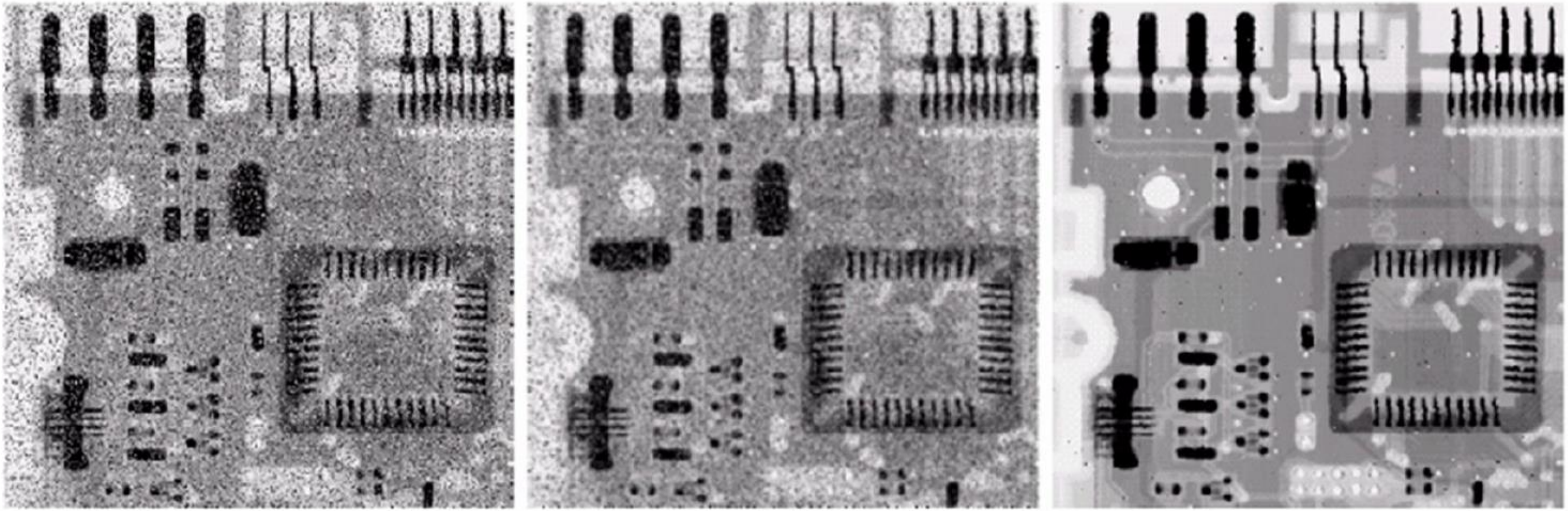
- 10, 15, 20, 20, 20, 20, 20, 25, 100

Non-linear filters → Order statistics filters → Median filter

- Median filtering is particularly effective in the presence of impulse noise (salt and pepper noise)
- Unlike average filtering, median filtering does not blur too much image details

- Advantages:
  - Removes impulsive noise
  - Preserves edges
- Disadvantages:
  - performance poor when # of noise pixels in the window is greater than 1/2 # in the window
  - performs poorly with Gaussian noise

Non-linear filters → Order statistics filters → Median filter



a) X-ray image of circuit board corrupted by salt-pepper noise
b) Noise reduction with a 3 x 3 averaging mask
c) Noise reduction with a 3 x 3 median filter

Blurring v/s Sharpening

- As we know that blurring can be done in spatial domain by pixel averaging in a neighbours
- Averaging is analogous to integration
- Therefore, logically it can be concluded that the sharpening must be accomplished by spatial differentiation

Derivative operator

- The strength of the response of a derivative operator is proportional to the degree of discontinuity of the image at the point at which the operator is applied.
- thus, image differentiation
  - enhances edges and other discontinuities (noise)
  - deemphasizes area with slowly varying gray-level values.

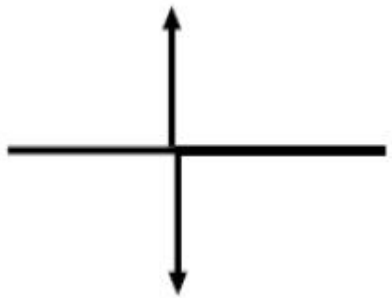
Derivative operator

- First-order derivatives generally produce thicker edges in an images
- Second-order derivatives have a stronger response to fine detail (e.g. thin lines or isolated points).
- First-order derivatives generally have a stronger response to a gray-level step
- Second-order derivatives produce a double response at step changes in gray level

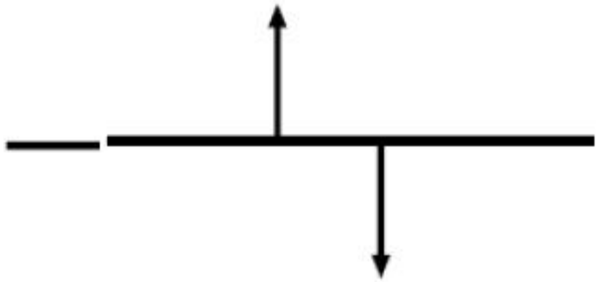
Sharpening Spatial Filters

- The principal objective of sharpening is to highlight fine detail in an image or to enhance detail that has been blurred
- First and second order derivatives are commonly used for sharpening

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$



$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



The Laplacian
(Use of Second derivative for Enhancement)

- The filter is expected to be isotropic: response of the filter is independent of the direction of discontinuities in an image.
- Simplest 2-D isotropic second order derivative is the Laplacian:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

2-Dimensional Laplacian

The digital implementation of the 2-Dimensional Laplacian is obtained by summing 2 components

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial x^2}$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

0	1	0
1	-4	1
0	1	0

Derivative operator → Laplacian (Second derivative)

0	1	0
1	-4	1
0	1	0

0	-1	0
-1	4	-1
0	-1	0

← 90° isotropic

45° isotropic →

1	1	1
1	-8	1
1	1	1

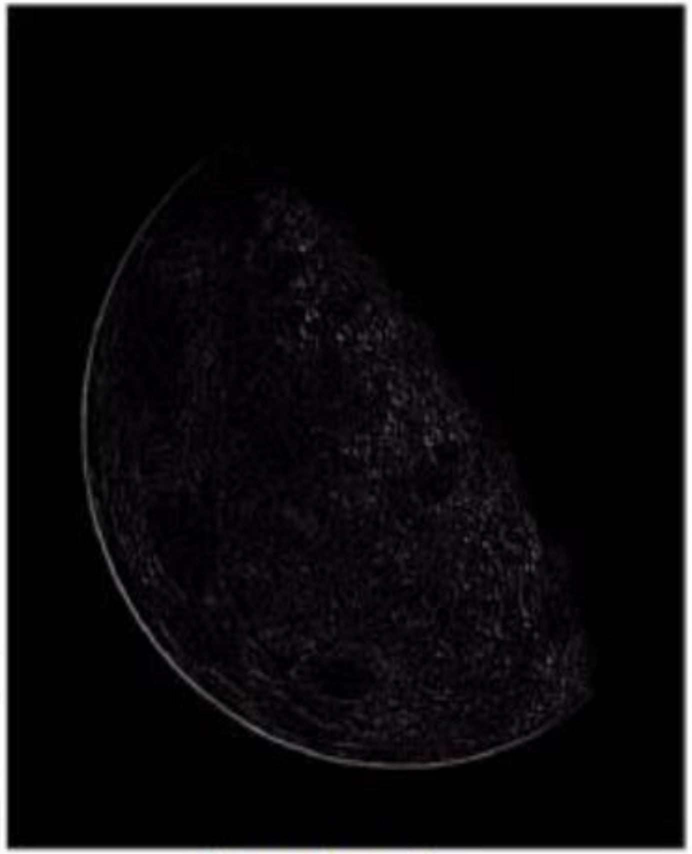
-1	-1	-1
-1	8	-1
-1	-1	-1

Derivative operator → Laplacian (Second derivative)

- As it is a derivative operator,
  - it highlights gray-level discontinuities in an image
  - it deemphasizes regions with slowly varying gray levels
- Tends to produce images that have
  - grayish edge lines and other discontinuities, all superimposed on a dark
  - featureless background

Derivative operator → Laplacian (Second derivative)

Original
Image



Laplacian
Filtered Image

Sharpened
Image

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Image of the north pole of the moon